

2.4 GHz Microstrip Patch Antenna with Inset Feed

Design, Simulation & Optimization Report

Ansys HFSS · Wi-Fi / Bluetooth ISM Band

1. Project Overview

This project focuses on the design, modelling, and optimisation of a **linear-polarised Rectangular Microstrip Patch Antenna** engineered to operate at the **2.4 GHz ISM band** (Wi-Fi / Bluetooth applications). Using *Ansys HFSS*, the patch geometry and microstrip inset feed were optimised to achieve maximum power transfer, high return loss, and a directional broadside radiation pattern.

2. Technical Specifications & Realised Data

Table 1: Design parameters: targets vs. simulated results.

Design Parameter	Target	Achieved (Sim.)	Physical Reason / Interpretation
Resonant Frequency f_0	2.40 GHz	2.3884 GHz	Minor downward shift ($\approx 0.48\%$) due to fringing fields expanding the electrical length beyond the physical patch boundaries.
Return Loss S_{11}	≤ -10 dB	-35.76 dB	Exceptional impedance matching at resonance; less than 0.03% of incident power is reflected back to the source.
VSWR at f_0	≤ 1.50	1.11 (at 2.388 GHz) / 1.26 (at 2.399 GHz)	Low VSWR values confirm an excellent match to the $50\ \Omega$ feed line, minimising reflection losses.
-10 dB Bandwidth	As high as possible	65 MHz (2.356–2.421 GHz)	Typical narrowband behaviour of microstrip patches ($\approx 2.7\%$ fractional bandwidth), dictated by the thin substrate structure.
Peak Gain	> 2.0 dBi	2.81 dBi	Broadside radiation characteristic suitable for localised wireless communication links.

3. Design Methodology & Intuition

3.1. Analytical Sizing — Transmission Line Model

The initial physical parameters of the patch were computed using standard microstrip design equations based on the substrate material characteristics.

Patch Width (W). Controls the radiation pattern and input resistance. Calculated to support efficient radiation fields along the edges:

$$W = \frac{c}{2f_0} \sqrt{\frac{2}{\epsilon_r + 1}} \quad (1)$$

Effective Dielectric Constant (ϵ_{eff}). Since the EM fields travel partly in the substrate and partly in the air, an effective dielectric constant accounts for this inhomogeneous medium:

$$\epsilon_{\text{eff}} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left[1 + 12 \frac{h}{W} \right]^{-1/2} \quad (2)$$

Patch Length (L). Determines the resonant frequency. Due to fringing fields at the patch edges, the antenna appears electrically longer than its physical length by $2\Delta L$. The actual length is:

$$L = \frac{c}{2f_0\sqrt{\epsilon_{\text{eff}}}} - 2\Delta L \quad (3)$$

3.2. Impedance Matching via Inset Feed

The input impedance at the radiating edge of a typical patch antenna is highly resistive (ranging between $200\ \Omega$ and $400\ \Omega$). To connect a standard $50\ \Omega$ microstrip transmission line directly without an external matching network, an **inset feed technique** was utilised.

As the microstrip line penetrates the patch by an inset depth y_0 , the input resistance drops rapidly following a cosine-squared behaviour:

$$R_{\text{in}}(y_0) = R_{\text{in}}(0) \cos^2\left(\frac{\pi}{L} y_0\right) \quad (4)$$

Through parametric sweeps of the inset length and gap width in HFSS, the feed position was fixed at the point where the input impedance tightly matches the $50\ \Omega$ characteristic line impedance.

4. Parametric Optimisation — Patch Width Sweep

4.1. Overview

To fine-tune the resonant frequency and impedance matching, a parametric sweep was performed over the patch width W , ranging from **28.5 mm** to **29.8 mm** in discrete steps, while all other dimensions were held constant. The S_{11} response was recorded across the frequency range 2.148–2.597 GHz for each width value.

4.2. Effect of Patch Width on Resonant Frequency

Increasing the patch width W shifts the resonant frequency downward. This behaviour arises because a wider patch increases the effective electrical length of the radiating structure, thereby lowering the frequency at which the patch resonates. The relationship follows from the width-dependent effective dielectric constant (Eq. 2): a larger W/h ratio increases ϵ_{eff} , which in turn reduces the resonant frequency.

Key observations from the sweep are summarised below:

Table 2: Resonant frequency and peak S_{11} as a function of patch width.

Patch Width W (mm)	Resonant Freq. (GHz)	Peak S_{11} (dB)
28.5	≈ 2.41	≈ -43
28.6	≈ 2.40	≈ -42
28.7	≈ 2.39	≈ -46
28.8	≈ 2.38	≈ -40
28.9	≈ 2.37	≈ -50
29.0	≈ 2.36	≈ -51
29.8	≈ 2.30	≈ -10

4.3. Width Sweep — S_{11} Plot

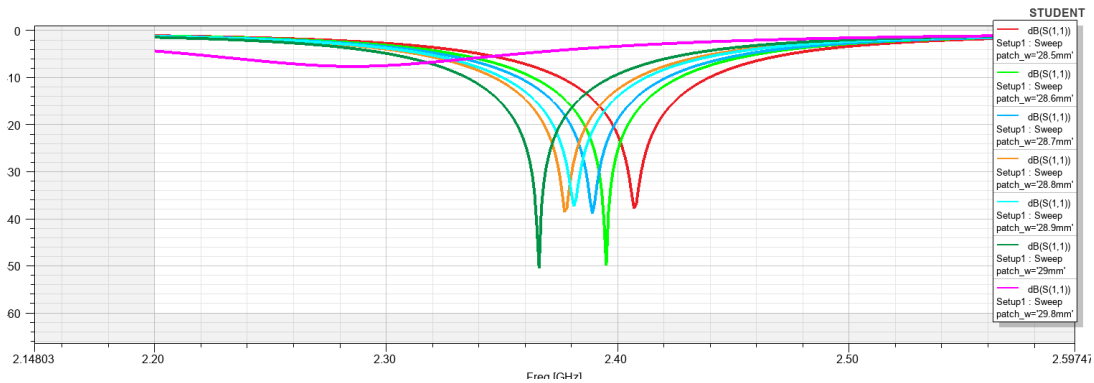


Figure 1: Parametric sweep of S_{11} (dB) vs. frequency for patch widths from 28.5 mm to 29.8 mm. Each curve corresponds to a single width value. As W increases, the resonance dip shifts toward lower frequencies. The deepest nulls (≈ -50 dB) occur near $W = 28.9$ – 29.0 mm, indicating optimal impedance matching in that range.

4.4. Design Insight

The sweep reveals a clear trade-off:

- Widths of **28.9–29.0 mm** yield the deepest S_{11} nulls (best matching), but push the resonance slightly below 2.4 GHz.

- A width of **28.6–28.7 mm** keeps the resonance closest to the target 2.4 GHz while still achieving a return loss well below -30 dB.
- At $W = 29.8$ mm the resonance has shifted so far down that the -10 dB bandwidth no longer covers the 2.4 GHz ISM band.

The final design adopts $W \approx 28.6$ mm as the optimal width, balancing frequency accuracy with impedance matching depth.

5. Simulation Results & Visualisations

5.1. Substrate Layout and Antenna Geometry

The structure incorporates a planar radiating element over a continuous ground plane, driven by a microstrip feedline recessed into the patch.

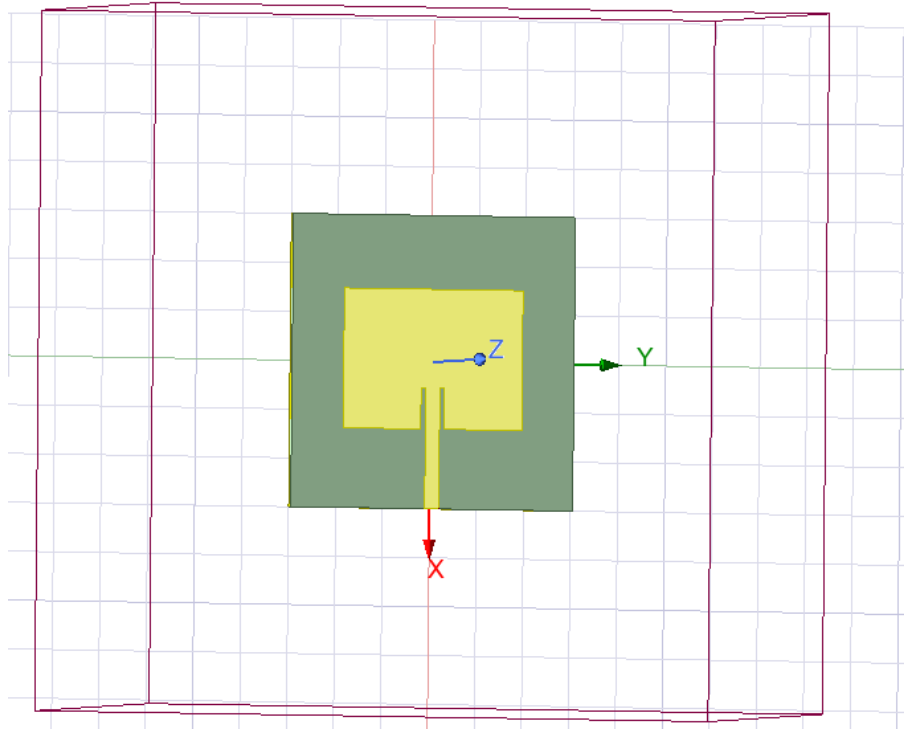


Figure 2: HFSS model of the rectangular microstrip patch antenna showing the patch, inset feed, and ground plane.

5.2. S -Parameter (S_{11}) Characterisation

The log-magnitude plot shows a deep resonance peak of -35.76 dB at 2.3884 GHz. The operational bandwidth determined at the standard -10 dB threshold spans from 2.356 GHz to 2.421 GHz, translating to an available bandwidth of **65 MHz**.

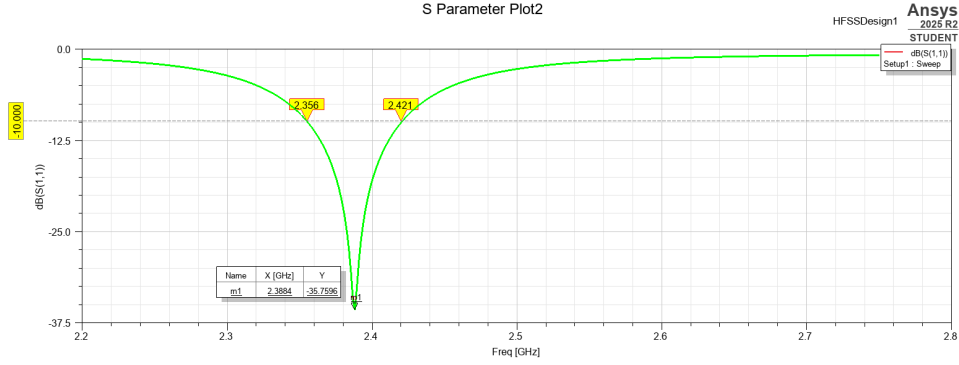


Figure 3: S_{11} log-magnitude response showing a resonance dip of -35.76 dB at 2.3884 GHz and a -10 dB bandwidth of 65 MHz (2.356 – 2.421 GHz).

5.3. Voltage Standing Wave Ratio (VSWR)

The VSWR curve drops into the ideal zone near the resonant point. At marker m_1 (2.3994 GHz), the VSWR is 1.26 . At the exact reflection dip (2.3884 GHz), the value converges near 1.11 , proving high power transfer into the radiation field.

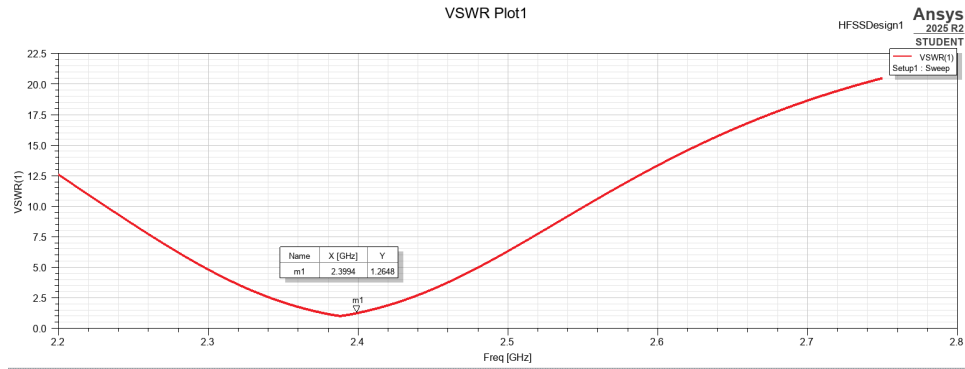


Figure 4: VSWR vs. frequency. The value at resonance approaches 1.11 , confirming near-perfect impedance matching to the $50\ \Omega$ feed line.

5.4. Input Impedance (Z -Parameters)

The magnitude profile of the input impedance $|Z(1,1)|$ tracks the parallel resonant behaviour of the patch. At 2.3994 GHz, the total magnitude is $63.14\ \Omega$. The peak shifts slightly higher than the S_{11} minimum because the complex reactive part passes through zero ($X_{in} = 0$) right at the true resonant point, balancing out the structure's capacitive and inductive fields.

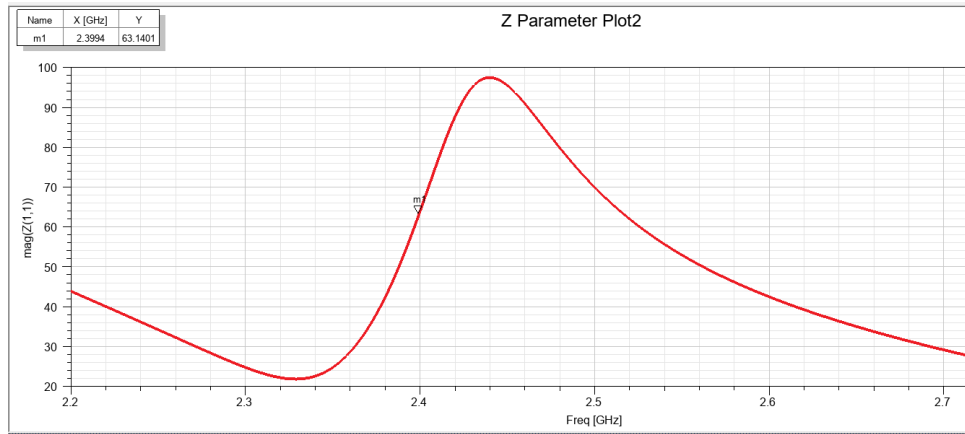


Figure 5: Input impedance magnitude $|Z(1,1)|$ vs. frequency. The peak near resonance reflects the parallel-resonant LC behaviour of the patch; the reactive part X_{in} crosses zero at the true resonant frequency.

5.5. 3D Radiation Pattern and Gain

The 3D far-field radiation plot confirms standard **broadside performance**. The antenna focuses radiation perpendicular to the patch surface (along the $+Z$ axis) with a peak total gain of **2.81 dBi**. The symmetric, lobe-free pattern indicates that surface wave losses are well-managed within the substrate.

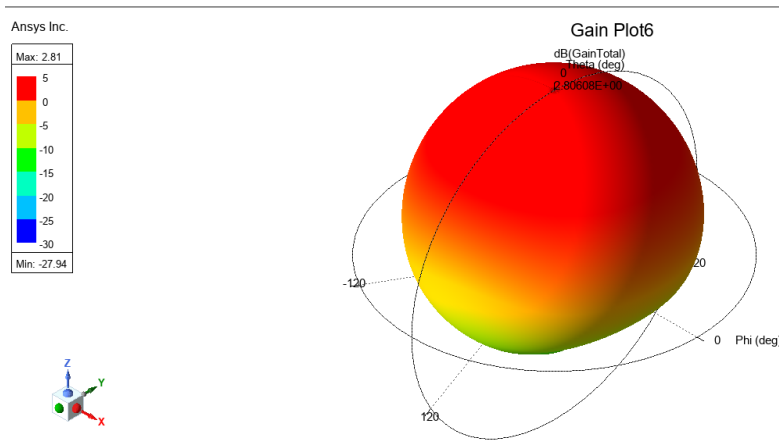


Figure 6: 3D far-field total gain radiation pattern at 2.3884 GHz. Peak gain of 2.81 dBi is directed along the broadside ($+Z$) axis, with a symmetric single-lobe profile.